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1 Appendix G: Active-Inference Behavioral Demo on IR Cues

1.1 Objective

Demonstrate a deterministic active-inference step for olfactory search under IR cues.

1.2 Implemented (stub) Methods (src)

- `src/behavioral_models.py`
 - `olfactory_active_inference_step(state, params)` — deterministic single-step update used in the demo

1.3 Script and Outputs

- Script: `scripts/generate_active_inference_demo.py`
- Data: `output/data/active_inference_demo.npz`
- Figure: `figures/active_inference_trajectory.png`

1.4 Figure

[width=0.9]figures/active_inference_trajectory.png

Figure 1: *Planned trajectories and belief updates for IR-guided search in a deterministic grid environment.*

1.5 Equation References

- Response/latency and information metrics: see `sec:mathematical_appendix`.

1.6 Reproducibility

- Run: `python3 scripts/generate_active_inference_demo.py`
- Artifacts saved to `output/data/` and `figures/`.
- Seed set to 42 via `src/config.set_random_seed(42)` for deterministic policy traces.
- Implementation note: the demo is a lightweight, deterministic adapter that calls `src/` policy utilities without embedding scientific logic in the script.

1.7 Cross-References

- Methods: `sec:methodology`
- Symbols: `sec:symbols_glossaryMathappendix` : `sec : mathematical_appendix`

[width=0.9]figures/active_inference_trajectory.png

Figure 2: *Deterministic active-inference trajectory generated by 'scripts/generate_active_inference_demo.py' using 'src/case_studies/olfactory_active_inference_step'. Figure and caption are repeated.*